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Tanouri, Afshin, Mulcahy, Rory, & [Russell-Bennett, Rebekah](#) (2019)

Transformative gamification services for social behavior brand equity: a hierarchical model.

Journal of Service Theory and Practice, 29(2), pp. 122-141.

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<https://doi.org/10.1108/JSTP-06-2018-0140>

Transformative gamification services for social behavior brand equity: A hierarchical model

Purpose: The first aim of this study was to test a hierarchical model with interrelationships between social support, value, and brand equity to examine the effect of a transformative gamification service on users' well-being behaviors. The second aim of the study was to demonstrate the usefulness of brand equity to measure social behavior brands encouraged via transformative gamification services.

Design/Methodology/Approach: Data were collected ($N=351$) via users of a transformative gamification service delivered via mobile in Iran. Structural equation modeling was used to test the model.

Findings: The results revealed the hierarchical model had a superior fit to the data than rival models measuring constructs at lower orders. The results also reveal value mediates the relationship between social support and brand equity for a social behavior.

Originality and Value: This is the first study to hierarchically test a model for transformative gamification services. Furthermore, it begins to shed light on the antecedents of value created within transformative gamification services, which to date has not been thoroughly explored. Finally, the study demonstrates brand equity is applicable beyond commercial campaigns and services and can be used to measure social (well-being) behaviors.

Keywords Brand equity, Gamification, Social behavior, Transformative services

Article Classification Research paper

1. Introduction

Health and well-being are central to the global economy and significant steps are being made by service scholars and practitioners to improve individuals' quality of life (Sweeney *et al.*, 2015). This focus has led to the growing area of transformative services – services which assist uplifting the well-being of customers (Anderson *et al.*, 2013). However, significant health issues remain worldwide, such as obesity resulting from overconsumption of food and beverage products (Seiders and Petty, 2004). The World Health Organization (WHO) reports that obesity has nearly doubled since 1975, with 1.9 billion adults being overweight and 650 million being classified as obese (WHO, 2017). This is concerning, because obesity is noted as leading to other health complications, such as diabetes, psychosocial problems, osteoarthritis, and cancer (Reilly and Kelly, 2011). Service scholars and service practitioners therefore still have an important role to play in delivering these services and understanding how they can address such health issues.

Traditionally, health and well-being customers are required to be physically present to receive services which improve their well-being (e.g. visiting a dietitian or doctor); however, in recent years, mobile technology has been increasingly used to deliver such transformative services (Schuster *et al.*, 2013, 2015). In addition to the growing prevalence of transformative services being delivered by technology, new approaches to how these experiences are delivered are emerging. One such approach is gamification – the use of game design elements which enable greater value-creation for service users (Huotari and Hamari, 2017). Service practitioners and researchers have realized a great opportunity exists in using “gameful” thinking for service provision, with the mobile gaming industry growing at 19% per year to a total of US\$46.1 billion in 2017 (McDonald, 2017). Further, transformative services and gamification sit at the nexus of two prevalent areas of service research: improving health and well-being through the delivery of services (Ostrom *et al.*, 2015), and the use of gamification

to encourage real-world behavior (Johnson *et al.*, 2016). Thus, it is timely to understand how gamification and transformative services can be used in tandem to improve customer well-being behaviors.

Transformative services are known to create an array of support resources (Rosenbaum and Massiah, 2007; Van Dolen and Weinberg, 2017) and value (Blocker and Barrios, 2015; Van Oerle *et al.*, 2016). However, little service literature exists to explain how support resources and value are created via gamified mobile applications or apps (hereafter referred to as “transformative gamification services”). This is despite the existence of such practices in the marketplace. For example, FitBit+ provides support via encouraging statements, “*Almost there! Get 1,487 steps more to hit your daily goal*”, and provides value to users via the tracking of physical activity and weight loss. Therefore, whilst it could be presumed support created by transformative gamified apps creates value, there is limited empirical evidence validating such an assumption. Furthermore, given the large array of different social support dimensions and value dimensions proposed in the service literature, it becomes hard to holistically understand their collective importance. Research suggests hierarchical modeling can be one useful approach to gain a greater holistic understanding of technological experiences (Atker *et al.*, 2011; Piyathasan *et al.*, 2015; Wetzels *et al.*, 2009). Thus, the first aim of this research is to empirically investigate the relationship between social support and value hierarchically in transformative gamification services.

Transformative gamification services can be unique in that they often encourage a social brand; that is, a behavior variant with attributes that bear similarities to that of a commercial brand. The notion of a social brand being a behavior is relatively new to the literature (Mulcahy *et al.*, 2015; Naidoo and Abratt, 2018), with the predominant alternative approach being social brands represented by campaigns or services (Evans *et al.*, 2002; Gordon *et al.*, 2016). This new and under-researched perspective of a social brand suggests

that behaviors can act and be measured as brand variants. For example, Mulcahy and colleagues' (2015) study suggests there are three variants of alcohol drinking behavior: absent drinking, moderate drinking, and binge drinking. Their study validates the performance of social behaviors being measured via brand equity. However, this is the only study to the authors' knowledge that validates and quantifies social behavior brands. Therefore, there is a need to confirm if indeed behaviors can function as a social brand. Further, what also requires examination is how social support and value created by gamified transformative services support a social behavior brand, and whether these effects are direct (i.e. social support to social brand equity) or mediated (i.e. social support to social brand equity via value). Thus, the second aim of this research is to investigate the effect of social support on social brand equity, and whether this effect is direct or mediated by value.

One context in which these aims can be examined is that of a transformative gamification service encouraging healthy beverage consumption outside of a western culture, such as by Middle Eastern consumers. This is an important area to research transformative gamification services, as research points out that many gamification studies are situated within a western context and focused on behaviors such as alcohol consumption (Dietrich *et al.*, 2018) or exercise (Johnson *et al.*, 2016). Further, industry reports show that the consumption of beverages that can be perceived as "unhealthy," such as soft drinks, has increased from 33.8 million liters in 2012 to 38.7 million in 2017 (Statista, 2018a). Therefore, it is important to clarify the generalizability of the positive impact of transformative gamification services in encouraging behaviors, such as healthy beverage consumption, beyond market segments situated in developed, western market economies.

The remainder of this paper is structured as follows. First, related literature is reviewed in the areas of gamification services, social support, value, and brand equity. This is followed by a presentation of the conceptual model and hypothesized relationships. The

method of the study is then outlined and the results presented. A discussion of the results and their theoretical and practical implications is then provided. The paper concludes by outlining directions for future research. Overall, this paper contributes new knowledge by testing a hierarchical model with mediating effects, which takes into consideration social support, value, and social brand equity in a new well-being behavior and market segment setting. This newly proposed theoretical model and empirical findings further the agenda of gamification delivered via mobile platforms (Hofacker *et al.*, 2016), and the service research priorities of Ostrom and colleagues (2015).

2. Literature review

2.1 Transformative gamification services

There is growing interest in the application of gamification in the services field. Research suggests gamification can create enjoyable and useful experiences (Koivisto and Hamari, 2014), which can lead to improved engagement (Harwood and Garry, 2015). In the literature, there appears to be two streams of research in services relating to gamification. The first stream focuses on gamification for commercial benefit; that is, examining how gamification can increase consumers' usage of products and services which are commercially orientated, such as e-banking services (Rodrigues *et al.*, 2016) or online shopping (Poncin, *et al.*, 2017). The second stream focuses on gamification for uplifting and maintaining change in consumers' performances of health and well-being behaviors (Hammedi *et al.*, 2017; Johnson *et al.*, 2016), which can be labelled "transformative gamification" (Mulcahy *et al.*, 2018). Most research in transformative gamification has focused upon health and well-being behaviors, such as exercise (Hamari and Koisito, 2015), smoking (Mulcahy *et al.*, 2018), and energy consumption (Johnson *et al.*, 2017). Furthermore, most studies are situated within developed economies and consumers from western markets such as the United States and

Australia. The current research, however, focuses on a less-researched behavior in transformative gamification services – healthy beverage consumption – with consumers from an emerging non-western economy, namely Iran. Taking such a perspective and approach is important to extending the knowledge base of transformative gamification services and its generalizable effectiveness across different contexts. The next section reviews the literature on social support to begin setting the conceptual foundations of the study's model.

2.2 Social support

In the literature, social support has been shown to be beneficial for assisting consumers to maintain or improve their well-being (Coulson, 2005; Rosenbaum and Massiah, 2007). This includes delivering social support through technological channels, such as online social support groups (Coulson, 2005; Parkinson *et al.*, 2017). Whilst not thoroughly researched, transformative gamification services can offer social support by providing guidance and encouragement to users. When conceptualizing social support there appears to be agreement that the construct should be measured as a multi-dimensional construct to allow richer insight into the construct's characteristics (Osman *et al.*, 2014; Yao *et al.*, 2015). Further, most studies appear to leverage the conceptualization of Cutrona and Russell (1990), who propose five dimensions: emotional, informational, esteem, network, and tangible assistance. However, despite agreement in the literature that social support is multi-dimensional and that many studies leverage the work of Cutrona and Russell (1990), the number and characteristics of social support differ considerably in different studies (see Table 1). For example, Rosenbaum and Massiah (2007) examine emotional and instrumental (informational) support, whereas Parkinson and colleagues (2017) measure all five dimensions of social support: emotional, informational, esteem, network, and tangible assistance. The variance in the number of social support dimensions examined across this study is to be expected, as it is dependent on the behavior, service user, and service under

investigation. Therefore, for the purposes of this research, social support is conceptualized as a two-dimensional hierarchical construct represented by emotional and informational support. These two dimensions were chosen as they represent the social support given by the transformative gamified service in addition to them being the most consistently measured dimensions in social support studies (see Table 1).

<Insert Table 1 about here>

2.3 Value creation

The creation of value is central to services, and particularly those that are transformative (Zainuddin *et al.*, 2016) or gamification-focused (Hamari and Kivisto, 2015; Leclercq *et al.*, 2017). In the literature, multiple multi-dimensional frameworks have been used to examine value created by technological products and services. Of these, three appear to be predominately used to guide studies of value: consumption-values theory (Sheth *et al.*, 1991), hedonic and utilitarian value (Babin *et al.*, 1994), and Holbrook's (1994) typology. Of these three, Babin and colleagues' (1994) and Holbrook's (1994) typology appear to be best suited to understanding game experiences and gamification. For example, Hamari and Koivisto (2015) conceptualize hedonic benefits to be represented by two single-factor dimensions – playfulness and enjoyment – and utilitarian benefits to be represented by usefulness and ease of use, whereas Okazaki (2008) leverages Holbrook's (1994) typology operationalized by Mathwick and colleagues (2001) to measure the value of mobile game experiences.

Research also suggests multiple frameworks of value can be used together to hierarchically measure and subsequently gain a deeper understanding of the concept's structure and how it collectively contributes to service users' experience (Wetzels *et al.*, 2009). Wetzels and colleagues' (2009) study, for example, demonstrates that value can be

measured hierarchically and this can contribute to understanding users' loyalty. They argue that measuring value hierarchically can be advantageous as it allows for examining the richness of the construct, its level of factors, and its overall assessment of impact on desired outcomes. They also demonstrate that hierarchically measuring value can enable multiple frameworks – such as Babin and colleagues' (1994) hedonic and utilitarian value, and Holbrook's (1994) typology – to be represented within a singular model. They use Holbrook's dimensions to measure experiential value and hierarchically model them according to Babin and colleagues' hedonic and utilitarian value dimensions. Research outside of value also demonstrates similar advantages for hierarchically modeling similar constructs, such as service quality in technological settings (Blut, 2016; Fassnacht and Koese, 2006; Piyathasanan *et al.*, 2015). However, research which goes beyond the single-order measurement of value and uses hierarchical modeling within transformative services and gamification is limited. This research takes a similar approach to Wetzels and colleagues (2009) by using Holbrook's (1994) typology to measure value, and Babin and colleagues' (1994) framework to hierarchically map their characteristics. Undertaking such an approach is important, as users are likely to evaluate certain dimensions of value created by a transformative gamified service, but also assess the cumulative value created by these dimensions. It is therefore proposed that hierarchical modelling can achieve these two purposes.

2.4 Brand equity and social brands

Brands can represent products, services, people, places, and ideas (Keller, 1998). Recently, branding has been extended to apply to social behaviors (e.g. drinking in moderation) (Mulcahy *et al.*, 2015). Keller (1998) points out that branding behaviors can be feasible and that actual behaviors need to be branded. In the literature, the predominate approach to

branding a social behavior relates to a supportive campaign or product/service (see Table 2). For example, the campaign ‘TruthSM’ is a brand which represents quitting smoking (Evans *et al.*, 2002), whereas services such as ‘Breastscreening Queensland’ studied by Gordon and colleagues (2016) can represent a brand for services which assist well-being through cancer screening. However, more recently a new and less examined approach has been proposed to measure social brands. Mulcahy and colleagues (2015) propose that branding frameworks, such as brand equity, can be applied to a social behavior without the co-branding of a campaign or supporting product or service. This bears similarities to other scholarly thoughts outside of branding, such as ‘value-in-behavior’ proposed by Gordon and colleagues (2018), which demonstrates that well-being behaviors themselves (without a product of service) can create value for consumers. Mulcahy and colleagues (2015) point out that there can be variants in the behaviors consumers can perform in a “behavior category”, which bear similarities to that of commercial product brands. For example, in their study they suggest that, like the commercial product of beer which can have brands with varying attributes (e.g. Heineken, Corona, and Peroni), so too can drinking behaviors (excessive drinking, moderate drinking, and abstinence drinking). They further demonstrate that the social brand of moderate drinking has similar brand equity dimension structures to that of commercial products. However, whilst Mulcahy and colleagues (2015) have successfully applied brand equity to social brands in alcohol consumption behavior and Australian adolescents, there is still a need to replicate and extend these findings to different social behaviors and market segments. It is important to extend the external validity (generalizability) of their findings and ensure they are applicable to different behavior contexts, such as healthy beverage consumption and Irian consumers (Kerr *et al.*, 2016). This study therefore examines the existence and structure of brand equity, specifically the dimensions of brand awareness, brand image, perceived quality, and brand loyalty, and how they contribute to the overall

brand equity (as a second higher-order factor, consistent with Blahut *et al.*, 2004) of a social behavior brand, and how this can be encouraged by the social support and value delivered via transformative gamification services.

<Insert Table 2 about here>

3. Conceptual model and hypotheses development

The literature review established the conceptual basis of the study and highlighted gaps in the literature. Based upon the gaps in the literature, this research presents a conceptual model (Figure 1) which hierarchically measures the constructs of social support, value, and brand equity. Furthermore, the model suggests there is a mediated relationship between social support and brand equity via value. Justification for the proposed model and its relationships are discussed below.

Based on prior research, it is hypothesized that social support created by transformative gamified services will create value for users. This follows the work of Loane and colleagues' (2015) netnographic study findings, which suggests that the social support generated from online health communities can create value for members. Similarly, Van Oerle and colleagues (2016) found online health communities that provide support and information to members can create affective (hedonic) and cognitive (utilitarian) value. Establishing the relationship between social support and value is important, as while prior studies have hinted at the link between these two constructs this has been mainly qualitative. Therefore, while there are suggestions that social support and value are linked, research is required to confirm these findings. Thus, based upon the evidence of prior studies, the following relationship is suggested for testing:

Hypothesis 1: Social support has a direct and positive association with value.

A review of the literature suggests alternative approaches regarding the interrelationships between social support, value, and brand equity. Rosenbaum and Massiah's (2007) study demonstrates that social support can have a direct effect on loyalty, a component of brand equity, whereas He and Li's (2010) study of high-tech services brand equity can be mediated by value. Research also suggests that aspects of brand equity can be antecedents rather than outcomes of customer value (Brodie *et al.*, 2009). As this study theorizes the impact of social support on creating value and support on social brand equity, it is appropriate and expected that social supports' impact will be mediated by value, and consequently have an indirect effect on brand equity. Thus, the following is proposed for testing:

Hypothesis 2: Social support has a significant indirect effect on brand equity via value.

Hypothesis 3: Value has a direct and positive association with brand equity.

<Insert Figure 1 about here>

4. Method

4.1 Study Setting

This study was conducted in Iran, a developing country in the Middle East with a population of over 80 million (Statista, 2018b). Conducting the study in Iran provided an interesting market segment to examine in relation to their technology adoption and usage, as well as excessive consumption of alcohol. First, in relation to the adoption of technology, industry

reports show that Iranian mobile users have reached 53 in 2018, an increase from 47 million in 2017 (Jafari, 2018). Thus, an opportunity exists for global and domestic technological service providers in Iran and the findings of this study besides its theoretical insights is how gamified technological services can be designed and marketed for this rapidly expanding market.

For the desired health and well-being behavior, reduced alcohol consumption, Iran also provides an interesting study setting. Despite alcohol being illegal in Iran, many alcoholic beverages are smuggled and distributed via the black-market or is illegally homemade (Lankarani and Afshari, 2014). Further, the social stigmatism and legal restrictions on alcohol consumption in Iran prevents consumers from admitting habits and problematic behaviours, which makes it difficult for hospital and medical services to identify and assist these consumers (Alavi, et al. 2016). Therefore, whilst excessive alcohol consumption is acknowledged as a health issue for Iran, its scale is somewhat difficult to determine and so too is the forecasting and planning of the delivery of services to overcome this issue. Transformative gamification services delivered via apps is one potential novel way to evade such difficulties and circumstances, as the use of mobile technology to gain access to health and medical services is known to provide greater anonymity and increase consumers' confidence in accessing potentially vital services in comparison to those which are face-to-face (Schuster, et al., 2013).

4.2 Data Collection and Sample

A survey was chosen as the data collection instrument for the study. Participants were recruited via social media posts, social media sites, and flyers with a download link to a purpose-built gamification app for the study (details regarding the app will be outlined shortly). Participants were incentivized to partake in the study with the chance to go into a

prize draw of a one-in-five chance to win a gold coin worth US\$100. Once participants had installed the app they were directed to a registry page with questions relating to their height, weight, age, and current habits regarding beverage consumption. This demographic information was important not only for the study, but it enabled the gamification app to tailor its information to the user. Participants were directed to use the gamified app for a period of 2 weeks, after which time they were invited to fill out a questionnaire with items relating to social support, value, and brand equity. The final sample consisted of 351 respondents. The sample profiles are shown in Table 3. The sample shows a good variability across gender and a good variation across those in normal weight (45.86%) and overweight or obese categories (50.99%) according to their self-reported body mass index (BMI), calculated using height and weight. The sample shows a slight tendency for respondents to be aged 20–35, which is to be expected as this group is known to adopt technology faster and perceive more value in the services it provides (Kumar and Lim, 2008).

<Insert Table 3 about here>

4.1 The gamified app

The participants used a gamified app which was developed by the lead author (see Figure 2). Gamification was incorporated into the app through game design elements, such as characters, competition, leaderboards, and points, all of which have been identified as key to services being “gamified” (Hamari *et al.*, 2014; Johnson *et al.*, 2016). The gamified app also included five mini-games. Users were rewarded points within the gamified app for performing desired behaviors in the real world (e.g. drinking a healthy beverage). This therefore facilitated the use of game design elements for purposes beyond entertainment and outside of the gaming system,

which is reflective of conceptualizations of gamification (Deterding *et al.*, 2011). The gamified app also provided messages of encouragement and informative facts regarding the beverages being consumed by users and the impact they had (positive or negative) on their health, reflecting emotional and informational support.

<Insert Figure 2 about here>

4.2 Instrument development

All measurements were based upon scales from previous studies. Items for emotional support were drawn from Uchida and colleagues (2008). Informational support consisted of items drawn from Oh and Lee (2012). Items for value were drawn from Mathwick and colleagues (2001), whereas brand equity items were drawn from Boo and colleagues (2009). All items were measured with 5-point Likert items. To ensure no biases resulted from the use of 5-point Likert scales, common method bias testing was conducted. A Harman's one-factor test and common latent factor test were carried out, as suggested by Podsakoff and colleagues (2003). The results revealed that common method bias did not contaminate the study's results.

5. Results

5.1 Instrument validation

Prior to testing the research hypotheses, the constructs were assessed for convergent and discriminant validity via confirmatory factor analysis. For the hierarchical analysis, social support and brand equity were measured as second-order hierarchical constructs, whereas

value was measured as a third-order hierarchical construct, which is consistent with prior studies (Wetzels *et al.*, 2009). Following the procedures suggested by Bagozzi and Yi (1988) and Fornell and Larker (1981), convergent validity was assessed via the *t*-values associated with the parameter estimates. Table 4 shows that the loadings of the items were positively and statistically significant for all constructs examined in the study. The reliability of the items was also confirmed, with the composite reliability index being higher than the recommended threshold of .60 (Fornell and Larker, 1981).

The constructs' hierarchical structures were also assessed. The two dimensions of support, emotional support (.833), and informational support (.945), loaded with significant *t*-values onto the hierarchical factor of social support, validating it as a second-order construct. Next, the hierarchical structure of value was assessed. Service excellence (.979) and Customer Return on Investment (CROI) (.974) had high factor loadings and significant *t*-values with the second-order construct of utilitarian value, confirming it as a second-order construct. Aesthetics (.860) and playfulness (.934) met factor-loading thresholds and had significant *t*-values, validating hedonic value as a second-order construct. Utilitarian value (.996) and hedonic value (.966) had high factor loadings and significant *t*-values, validating value as a third-order construct. Finally, brand equity's hierarchical structure was assessed, with brand awareness (.962), brand image (.989), perceived quality (.957), and brand loyalty (.990) having high factor loadings and significant *t*-values, confirming it as a second-order hierarchical construct. Discriminant validity was also established with AVE scores being above their related shared variances with constructs (see Table 5). After establishing the hierarchical structure of the constructs, the measurement model was assessed with the results (CMIN/DF=1.39 CFI=.97, TLI=.97, RMSEA=.03) suggesting the model had an acceptable fit to the data. The CMIN/DF ratio was smaller than the recommended 3 by Fornell and Larker (1981), indicating a good model fit.

<Insert Tables 4 and 5 about here>

5.2 Testing of hypotheses

To test the hypothesized relationships structural equation modelling in AMOS24.0 was performed. The structural model had a good fit to the data (CMIN/DF=1.39, CFI=.97, TLI=.97, RMSEA=.03) and the hypotheses testing results are detailed in Table 6 and Figure 3. Testing revealed social support had a positive association with value ($\beta=.38, p<.000$), as hypothesized in H1. This indicates social support creates a valuable transformative gamification service experience for users. H2 predicted social support would have a significant indirect effect on brand equity via value. The results revealed the direct effect of social support on brand equity was non-significant ($\beta=-.03, ns$), whereas results revealed (as predicted in H2) that social support had a significant indirect effect on brand equity via the mediator of value ($\beta=.34, p<.000$). The results also revealed value had a strong significant and direct positive association with brand equity ($\beta=.94, p<.000$), supporting H3. This suggests value created via transformative gamification services positively builds brand equity relating to the social behavior being encouraged. The model also produced a high R^2 of .87 for brand equity, suggesting the model was a good predictor.

<Insert Table 6 and Figure 3 about here>

5.3 Post-hoc analyses and rival models

This study took the a priori belief that hierarchically modeling the relationships between social support, value, and brand equity would be superior to lower-order modeling. It is also agreed in the literature that researchers should compare the fit of rival (competing) models (Iacobucci, 2010). Accordingly, alternative lower-order models were tested to indeed confirm that hierarchically modeling the constructs of support, value, and brand equity is better served to the purpose of understanding transformative gamification services and future research.

The first rival model tested examined all constructs as single-order factors, with relationships between the constructs following that of the proposed model, whereas the second model tested support as a single-order construct, value as a second-order construct, with playfulness and aesthetics representing hedonic value, CROI and service excellence representing utilitarian value, and brand equity measured via single-order factors. The resulting test statistics for the first rival model (CMIN/DF=14.07, CFI=.92, TLI=.75, RMSEA=0.194) and second rival model (CMIN/DF=2.45, CFI=.91, TLI=.90, RMSEA 0.06) were either not acceptable based upon recommended fit indices for structural equation modeling or inferior to the originally proposed and tested model (see Table 7). Thus, it is suggested that the data indicate the originally proposed and tested model is superior and more appropriate for theoretically and empirically understanding transformative gamification services.

<Insert Table 7 about here>

6. Discussion

This research investigates the influence of social support on value and social brand equity in a transformative gamification setting through the empirical testing of a hierarchical model.

The hierarchical model was shown to outperform rival models measuring constructs at a

lower order, and the effect of social support being fully mediated by value. These findings provide important theoretical and managerial contributions which will now be discussed.

6.1 Theoretical implications

This newly proposed theoretical model and empirical findings further the agenda of gamification delivered via mobile platforms (Hofacker *et al.*, 2016), and the service research priorities of Ostrom and colleagues (2015). Collectively, the results also both support and build on extant literature to provide new insight into the emerging area of transformative gamification in three ways.

6.1.1 Hierarchical Modelling for Gamification Service Research

The first theoretical contribution of this study relates to its hierarchical modelling approach. The results of this research indicate that the conceptual model fits well to the data and outperforms other competing models measured at lower orders. To the authors' knowledge, this study is one of the first to theoretically propose and empirically demonstrate a hierarchical model for transformative gamification. Previous modeling of gamification, including those of a transformative and commercial focus, have predominately modelled constructs such as value as single-order constructs (Koivisto and Hamari, 2014; Hamari and Kovisto, 2015). This research demonstrates an alternative theoretical view of conceptualizing and empirically testing related constructs in gamification hierarchically to provide a more holistic perspective of users' experiences (Piyathananan *et al.*, 2015, Wetzels *et al.*, 2009). Consistent with the emerging evidence in digital technologies and experiences, the findings of this study suggest value has a hierarchical structure in a transformative gamification setting (Wetzels *et al.*, 2009; Mencarelli and Lombart, 2017). Furthermore, our study extends this perspective by confirming that social support and brand equity can also

have hierarchical structures, which to date has not been thoroughly explored or validated in the literature.

Taking a hierarchical approach to modelling consumers experiences with transformative gamification is conceptually interesting and valuable for a number of reasons. First, it represents a step toward better understanding the associations between various sub-dimensions (for example the interrelationships and structure of value) (Faasnacht & Koese, 2006). Second, the findings of this study suggest hierarchical construct models provide utility to gamification and service scholars in that it allows for greater theoretical parsimony and a reduction in model complexity (i.e. the number of hypothesized relationships tested) (Wetzels, Odekerken-Schröder, Van Oppen, 2009). This is important for service researchers examining gamification in particular as many of the constructs and frameworks are multi-dimensional and often have a considerable amount of relationships between single order factor constructs. As a result, exciting opportunities exist for service and gamification scholars to test prior (and future) studies frameworks which theorize single order factors are related but yet to be empirically supported from a hierarchical modelling approach. Furthermore, this offers service and gamification some flexibility in their theorization and analysis by allowing diagnostic examinations of the effect of single order factors but also the cumulative effect of concepts.

6.1.2 Social Support, Value and Social Brand Equity Interrelationships

The second contribution of this study relates to a greater understanding of the interrelationships between social support, value and social brand equity. In this study, the findings show value plays a mediating role between social support and social behavior brand equity. Prior research has suggested that social support (Parkinson et al., 2017; Rosenbaum and Massiah, 2007) and value (Hamari and Kivitsa, 2015; Zainuddin et al., 2013) can

influence well-being outcomes. Yet, to the best of the authors' knowledge, this study is one of the first to show the strong positive influence social support has on creating value in transformative gamification. Consistent with prior research, this study found that value plays a mediating role (e.g. He and Li, 2010), and that social support has a non-significant direct effect on brand equity, but a significant indirect effect via value. This has potentially important implications beyond transformative gamification, as social support and value are two central concepts in the broader transformative service research agenda (Anderson et al., 2013). In addition, the gamification literature has established that value can influence desired outcomes (e.g. Hamari and Kivitsa, 2015), but the antecedents which influence the creation of value have not yet been thoroughly researched. This research therefore sheds light on social support as one form of antecedent that can result in value creation in transformative gamification services.

6.1.3 Behaviors as Social Brands

The third contribution of this study relates to measuring brands in transformative service and social marketing settings. Theorists in social marketing have posited that behaviors function as products and brands (Gordon, 2012). Yet, there is limited empirical evidence that supports this view, with most studies examining branding of services and campaigns which encourage a desired behavior (Evans *et al.*, 2002; Asbury *et al.*, 2008; Gordon *et al.*, 2016; Evans *et al.*, 2016). Findings from this research provide important empirical evidence that social behaviors can indeed be viewed as products measured via brands. In doing so, it theoretically extends the generalizability of prior studies – those prior studies which have thus far only provided preliminary evidence (Mulcahy *et al.*, 2015; Naidoo and Abratt, 2018). Therefore, this research strengthens the theoretical perspective that there are two approaches to branding for social behaviors: branding of the service or campaign, and branding of the social behavior.

6.2 Practical implications

There are several important practical implications derived from this study. At a higher level, the results of this study add to the evidence-base of transformative gamification assisting the well-being of consumers. As well-being and healthcare services are often funded publicly via government bodies or non-government organizations (NGOs), the results of this study provide further compelling evidence to governments, NGOs, and other well-being service providers to support the funding and creation of transformative gamification apps that target health and well-being behaviors. This could also be important for countries where the discussion of particular well-being behaviors with health professionals is a taboo topic or illegal such as the consumption of alcohol in countries such as Iran. The findings of this study therefore suggest that transformative gamification is one alternative that provides a potentially more anonymous approach for consumers to seek health and well-being services.

Our findings also reveal the importance of designing social support mechanisms to create valuable transformative gamification service experiences. Thus, transformative gamification designers should pay special attention to incorporating elements which provide social support to users. Specifically, from our conceptualization of social support, emotional support and informational support are encouraged to be especially designed within gamification transformative services. Emotional support should focus on boosting users' self-esteem and ego. For example, apps in the current marketplace such as *7 Cups: Anxiety & Stress Chat*¹ provides users with anonymous emotional support and counseling from trained professionals. Practitioners should consider ways in which design features such as chatrooms or communities incorporated into transformative gamification services to foster the seeking

¹ <https://itunes.apple.com/us/app/7-cups-anxiety-stress-chat/id921814681?mt=8>

and provision of (anonymous) emotional support. For informational support, the findings suggest that design features such as push notifications or tracking features can be designed within transformative gamified apps to create value for users. Practitioners could therefore undertake similar design approaches to apps such as *Strava*², which allows users to track their running or riding and analyse their performance overtime.

The model and findings from this study also demonstrate service and gamification practitioners are able to measure social support, value and social brand equity at multiple levels. Just as the hierarchical model approach provides flexibility for scholars, this approach also provides practitioners the ability to evaluate these constructs at a primary dimensional level (single order factor) or cumulative level (higher order). Practitioners can therefore make informed strategic decisions when evaluating and assessing gamification services based upon their evaluation aims as well as their information and reporting requirements (Dagger, et al. 2007). Finally, our findings show transformative service practitioners can measure well-being behaviors using branding models and frameworks such as brand equity (Mulcahy *et al.*, 2015). Practitioners could thus pay attention to the variation of the behavior they are encouraging customers to perform and pinpoint the defining characteristics or value propositions the behavior has in comparison to other competing behavior brands. Further, this demonstrates a new perspective and method by which transformative service practitioners can measure well-being. It is suggested that practitioners measure social brand equity for both their targeted well-being behavior and other competing behaviors to diagnostically evaluate the performance of their services and campaigns in encouraging well-being behavior.

6.3 Limitations and further research directions

² <https://itunes.apple.com/au/app/strava-running-and-cycling-gps/id426826309?mt=8>

Some limitations of this research will now be acknowledged. This study utilized a cross-sectional design of consumers' experiences after using a transformative gamification service for a period of two weeks. It would be interesting for future research to track consumers' experiences with transformative gamification services over time and identify how value and brand equity in a social behavior evolve (or diminish). A worthy direction for future research would be to consider other aspects aside from social support which assist in creating valuable transformative gamification service experiences. This could include the influence of game design elements, such as points, challenges, and characters, on the creation and experience of value. Caution should be taken when attempting to draw generalizability from the study's findings to other market settings and behaviors. Whilst this study examined transformative gamification services in a market segment (Iran) and behavior not thoroughly researched, future research may benefit from examining cross-cultural differences or variances which may occur based upon different behaviors being encouraged. Future studies could also seek to investigate whether the findings of the current study are consistent in other developing markets beyond Iran such as countries in the Middle East (e.g. Saudi Arabia and Qatar) and Latina America (for example Chile and Argentina). Another interesting avenue for future research would be to examine factors which moderate (e.g. strengthen or weaken) the relationships examined within this study. For example, whether consumers' previous experience or familiarity with technology (in the case of this research apps or smartphones) or duration of use based upon data analytics changes the nature of the relationships between social support, value and social brand equity. Finally, whilst outside the scope of this study, including some emerging concepts in transformative services and gamification could provide greater insight. For example, consumer engagement is emerging as an important concept for understanding value co-creation, gamification and services (Harwood and Garry, 2015;

Hollebeek and Andreaseen, 2018), and its incorporation in future studies could provide an interesting extension on the current study.

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Table 1.

Chronological overview of social support studies using Cutrona and Russell's (1990) typology

Author(s)/Year	Information	Tangible Assistance	Esteem	Emotional	Network
Coulson (2005)	Y	Y	Y	Y	Y
Rosenbaum and Massiah (2007)	Y			Y	
Coulson and Knibb (2007)	Y	Y	Y	Y	Y
Lin <i>et al.</i> (2015)	Y	Y	Y	Y	Y
Loane <i>et al.</i> (2014)	Y		Y	Y	Y
Mo and Coulson (2008)	Y	Y	Y	Y	Y
Oh <i>et al.</i> (2013)		Y	Y	Y	Y
Tong <i>et al.</i> (2013)	Y	Y	Y	Y	Y
Yao <i>et al.</i> (2015)	Y			Y	
Wang <i>et al.</i> (2015)	Y			Y	
Totals	9	6	7	10	7

Table 2.

Chronological overview of branding studies in social marketing

Author/s (Year)	Target Behavior	Branding of Behavior	Brand for Service/Campaign Assisting Behavior
Evans <i>et al.</i> (2002)	Smoking		Y
Evans, <i>et al.</i> (2007)	Smoking		Y
Asbury <i>et al.</i> (2008)	Physical Activity		Y
Price <i>et al.</i> (2009)	Physical Activity		Y
Lee and Hecht (2011)	Substance Use		Y
Mulcahy <i>et al.</i> (2015)	Alcohol Consumption	Y	
Gordon <i>et al.</i> (2016)	Cancer Screening		Y
Evans <i>et al.</i> (2016)	Smoking		Y
This Research	Beverage consumption	Y	

Table 3.

Respondents characteristics (%)

Demographic Variable	%
Gender	
Male	49
Female	51
Age	
<20	13.68
20–35	50.72
36–50	34.47
>50	1.13
BMI	
<18.5 (Underweight)	3.13
18.5–24.9 (Normal weight)	45.86
25–29.9 (Overweight)	38.74
>30 (Obese)	12.25

Table 4.

Construct reliability and validity

Construct	Loading (t-stat)	AVE	CR
<i>Emotional Support</i>		0.58	0.94
The app cheers you up when you are depressed.	0.78 (15.22***)		
The app cares about your health and shape, when you enter your weight and health status in the app.	0.78 (15.31***)		
When something good happens to your health and weight the app praises you and it seems to be happy.	0.75 (14.52***)		
If you fall into a health situation that you cannot do anything about, the app sees to it that something is done.	0.74 (14.31***)		
If the app finds out that you have failed to reach your health and weight target, it tries to console you.	0.75 (14.52***)		
If you are feeling down about your health and weight, the app cares for you right away.	0.73 (14.17***)		
If you make a mistake, the app picks up the slack for you.	0.77 (15.02***)		
If you accomplish something, the app congratulates you with sincerity.	0.77 (14.97***)		
If you are busy and you cannot finish your health and weight target, it helps you out.	0.77 (15.12***)		
The app recognizes your ability and it expects you to do the best of your abilities on a daily basis.	0.75 (14.63***)		
If the app finds out that you are having a hard time with your health and weight problems, it gives you advice on how to solve the problem.	0.76 (14.72***)		
The app accepts you for who you are, regardless of your health and weight problems.	0.74 (14.97***)		
<i>Informational Support</i>		0.60	0.85
The app offers you necessary information when you face weight-related problems.	0.77 (14.43***)		
The app gives you additional information about weight-related symptoms and treatments that you are not familiar with.	0.76 (14.22***)		
The app gives you advice so that you understand the reality of and adapt yourself to a healthy plan.	0.75 (14.10***)		
The app will show you how to cope with a difficult weight-related situation that you have faced.	0.77 (14.22***)		
<i>Playfulness</i>		0.60	0.74
Using X gets me away from it all.	0.88 (13.26***)		
Using X makes me feel like I am in another world.	0.65 (13.26***)		
<i>Service Excellence</i>		0.65	0.79
When I think of X I think of excellence.	0.73 (16.43***)		
I think of X as an expert in the content it offers.	0.87 (16.43***)		
<i>CROI</i>		0.71	0.83
Using X is an efficient way to use my leisure time.	0.86 (19.23***)		

Construct	Loading (t-stat)	AVE	CR
Using X fits in with my schedule.	0.81 (19.23***)		
<i>Aesthetics</i>		0.59	0.85
I like the way X looks.	0.68 (14.06***)		
X is visually appealing.	0.76 (16.43***)		
The way X displays content is attractive.	0.75 (16.04***)		
I think X is very entertaining.	0.85 (16.04***)		
<i>Brand Awareness</i>		0.71	0.82
[Behavior] is well known.	0.83 (19.42***)		
[Behavior] has a good name and reputation.	0.84 (19.42***)		
<i>Brand Image</i>		0.69	0.90
The image of [behavior] is consistent with my image of myself.	0.84 (21.73***)		
[Behavior] reflects who I am.	0.87 (21.73***)		
My friends would think highly of me if [I performed behavior].	0.76 (17.38***)		
[Behavior] reflects my personality.	0.84 (20.32***)		
<i>Perceived quality</i>		0.77	0.87
[Behavior] provides quality experiences.	0.87 (23.13***)		
[Behavior] provides experiences of consistent quality.	0.88 (23.13***)		
<i>Brand Loyalty</i>		0.73	0.88
I would advise my friends [to perform behavior].	0.83 (21.01***)		
[Behavior] would be my preferred choice of behavior.	0.87 (21.02***)		
Overall, I am committed to [behavior].	0.84 (20.13***)		

Note: Items in the table are translated into English. Items used in the study were in Persian, one of the main languages spoken in Iran where participants were drawn from. X=indicates placing for transformative gamification service name.

Table 5.Mean, standard deviation (*SD*) and correlations of constructs (higher order)

Construct	Mean	<i>SD</i>	1	2	3
1. Support	3.49	0.67	(.82)	.09	.08
2. Value	3.72	0.84	0.31**	(.96)	.67
3. Brand Equity	3.76	0.99	0.29**	0.82**	(.94)

AVE scores in brackets. ** $p < .01$, shared variances shown in top half of matrix

Table 6.

Structural model results

Paths	Direct effects	Indirect effects (mediated)
Support→Value	$\beta=0.38^{***}$	Same as direct
Support→Brand Equity	$\beta=-0.03^{ns}$	$\beta=0.36^{***}$
Value→Brand Equity	$\beta=0.94^{***}$	Same as direct
R^2		
Value	0.14	
Brand Equity	0.87	

* $p<.05$, ** $p<.01$, *** $p<.000$ *ns* non-significant

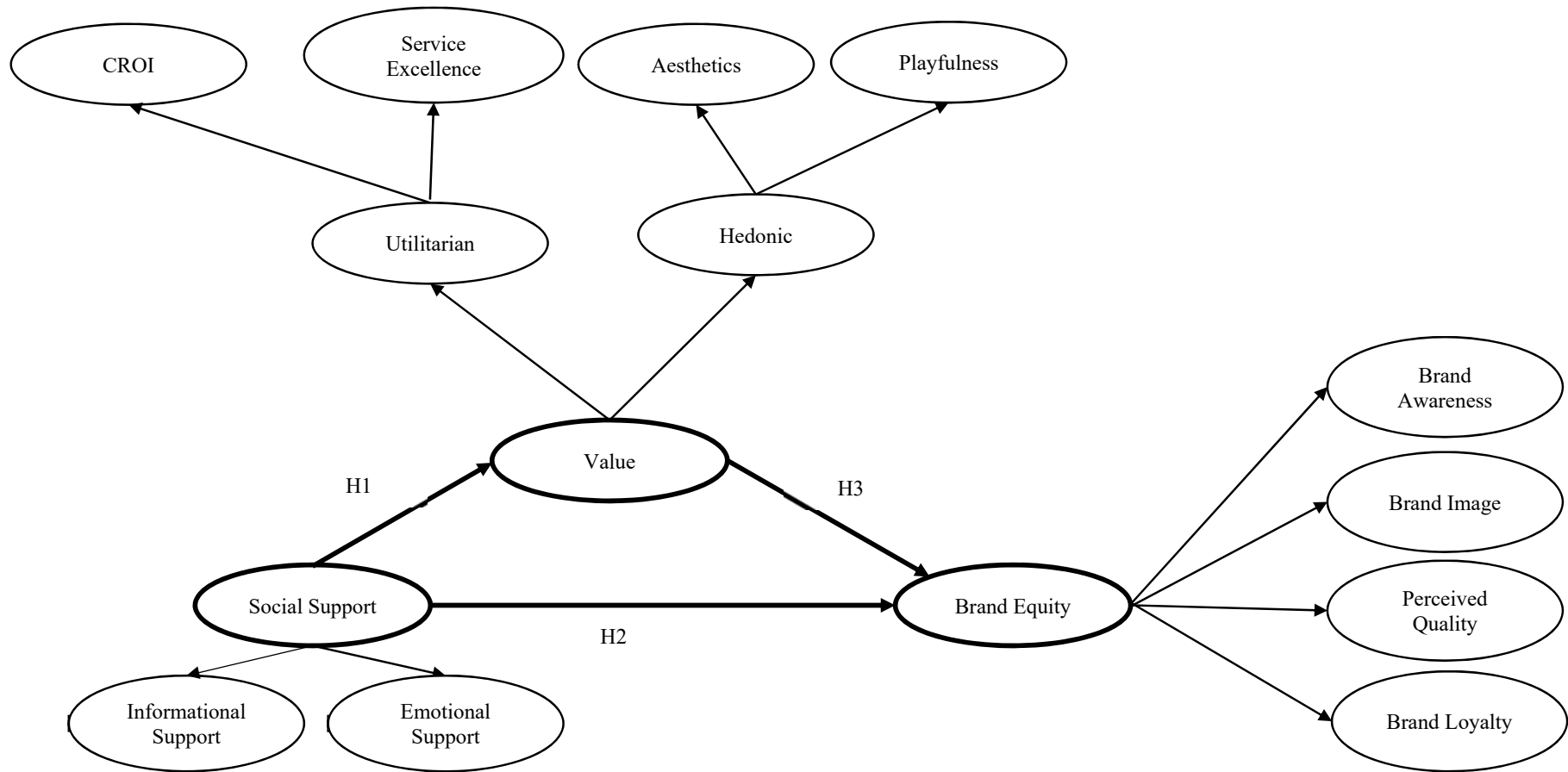
Table 7.

Comparison of fit indices for rival models and tested model

Fit Indices	Recommended Thresholds	Rival Model 1	Rival Model 2	Tested Model
CMIN/DF	<3.00	14.07	2.45	1.39
CFI	>.95	0.92	0.90	0.97
TLI	>.95	0.75	0.90	0.97
RMSEA	<.06	0.19	0.06	0.03

Note: Thresholds based upon Fornell and Larker (1981) and Iacobucci (2010).

Figure 1.
Conceptual model

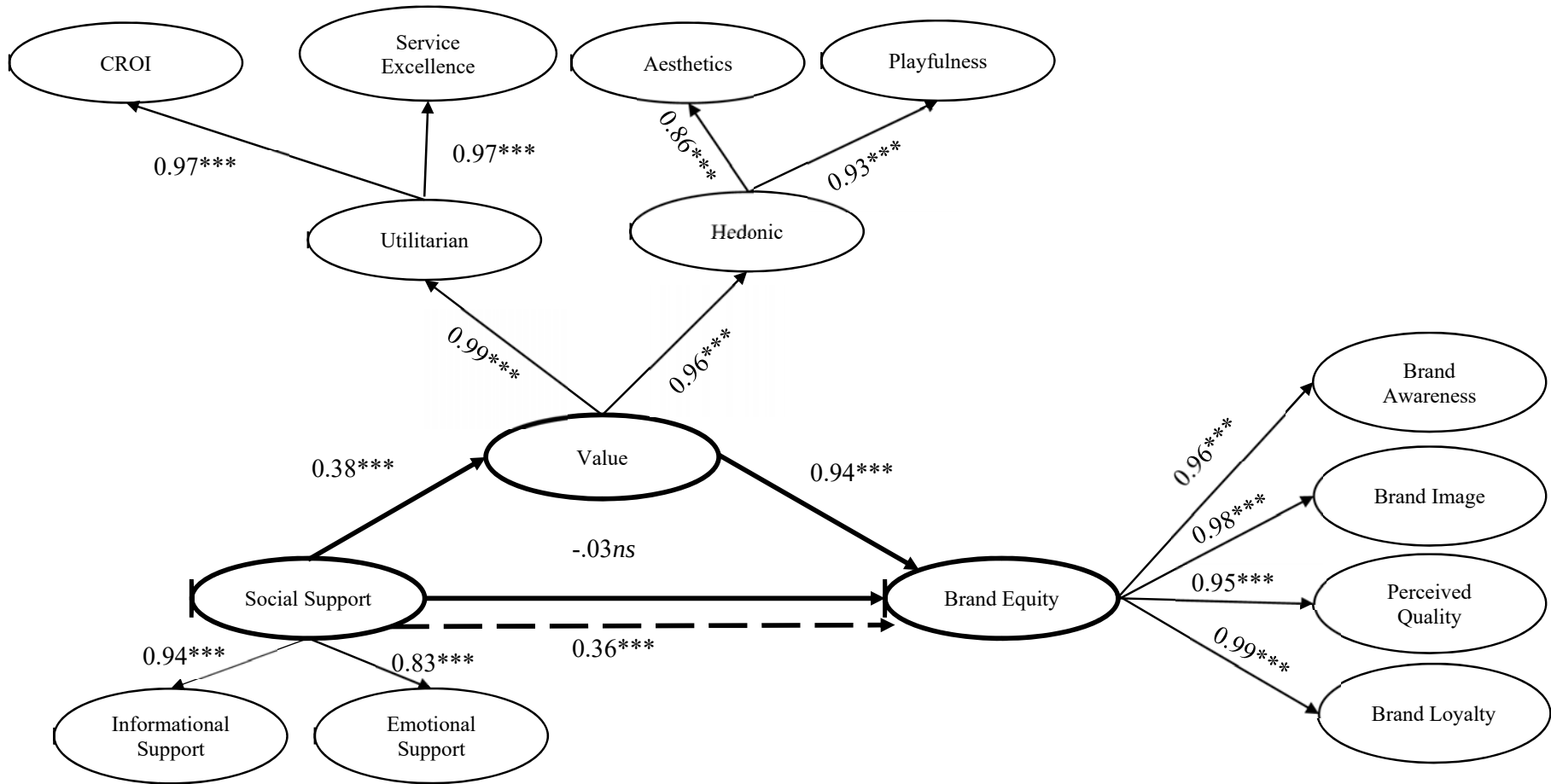


Notes: The hierarchical model for support, value, and brand equity is of the reflective–reflective type, which is suggested as superior to formative approaches (see Iacobucci, 2010).

Figure 2.
Screenshots of the gamified app used in the study



Figure 3.
Relationship testing results



Note: Dashed line indicates indirect effect, *** $p < .000$, ns=non-significant

